

WASP-50b

R-1375

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_151105 · created 2026-08-01T01:22:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457331.919 +/- 0.055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.02 +/- 1.76 [%]
Orbital Inclination (inc)	89.35 +/- 2.93
Airmass coefficient 1 (a1)	84.89 +/- 126.24
Airmass coefficient 2 (a2)	-2.95 +/- 0.35
Scatter in the residuals of the lightcurve fit is	73.99 %
Best Comparison Star	None
Optimal Aperture	1.56
Optimal Annulus	5.29
Transit Duration (day)	0.068 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.051 vs 0.1404 (deviation 0.61σ)
Duration fit vs published	0.068 d vs 0.0754 d (deviation 0.24σ)
Mid-time O–C	53.4 min
Depth SNR	3.2
Residual scatter	73.99 %

Light curve

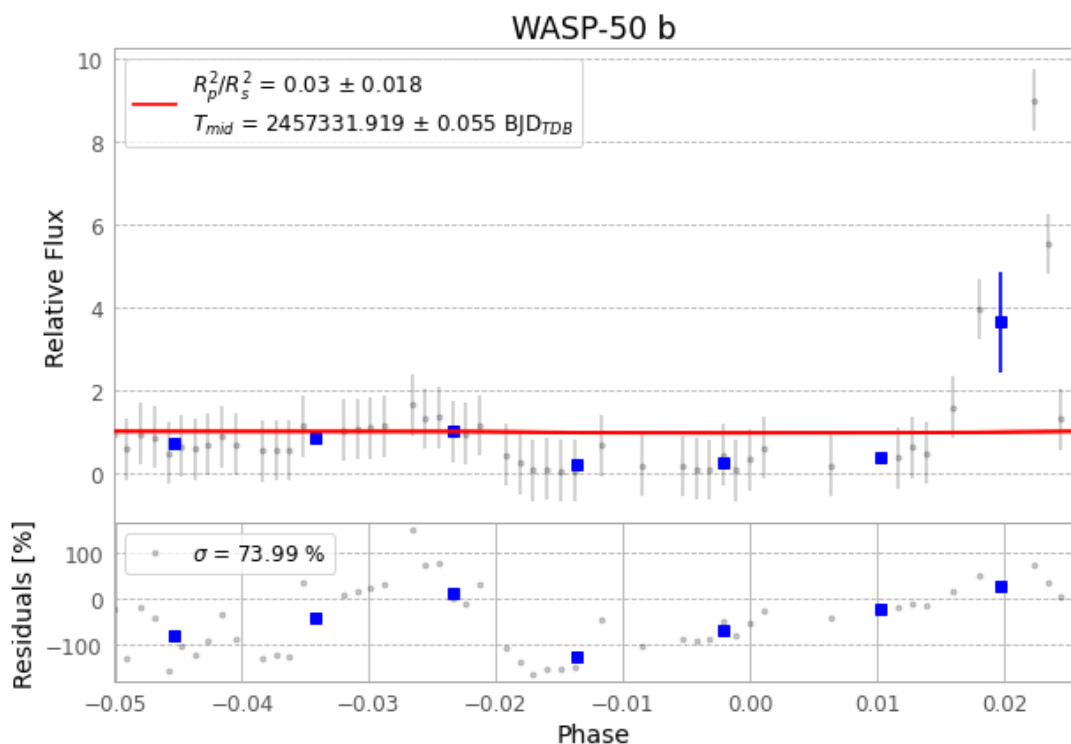


Figure 1 — FinalLightCurve_WASP-50 b_2015-11-05.png (SHA-256 6029669546e74ecc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7ff6c925bb6c4477...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

71 FITS frames (47 MB), WASP-50151105073312.FITS → WASP-50151105110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151105.log (f3548e209712...), FinalLightCurve_WASP-50 b_2015-11-05.png (6029669546e7...), FinalLightCurve_WASP-50 b_2015-11-05.csv (62fba6831b4...)

Compute resources

Wall time 230.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:22Z.